FED. SUP CLASS
2620

THE TIRE SHALL BE IN ACCORDANCE WITH THE APPLICABLE REQUIREMENTS OF SPECIFICATION MIL-R-7726 EXCEPT AS SPECIFIED HEREIN

SIZE	PLY RATING	STATIC LOAD RATING LBS.	INFL. PRESS. PSI RATED	BURST PRESS. PSI MIN.	BEAD WIDTH IN. MAX.	WEIGHT LBS. MAX.	STATIC UNBAL. ± 1/2 OZ.	TREAD	MOLD SKID DEPTH MIN.	DEFLC. + 2% - 4%
20.00-20	1/22	38,500	95	333	2/3.375	235	0	3/RIB	.40	32%

1/TIRE SHALL BE CAPABLE OF OPERATING AS A TUBELESS OR TUBE TYPE TIRE.
2/DOES NOT INCLUDE MAXIMUM BEAD TOE FLASH OF 3/8 INCH.
3/AT LEAST FOUR GROOVES SHALL BE WITHIN THE CONTACT AREA OF THE STATIC LOAD FOOTPRINT TAKEN AT 60% OF THE STATIC LOAD AND RATED INFLATION PRESSURE. TREAD MAY BE FABRIC REINFORCED IF NECESSARY

RIM DATA:

WIDTH BETWEEN FLANGES (INCH)	LEDGE DIAMETER (INCH)	FLANGE HEIGHT (INCH)	MIN. EDGE WIDTH (INCH)	MIN. FLANGE WIDTH (INCH)	FLANGE RADIUS (INCH)	HEEL RADIUS (INCH)	FLANGE EDGE RADIUS (INCH)
15.50	20.00	2.00	6.00	1.50	.875	.313	.125

THE TIRE COVERED BY THIS DRAWING SHALL BE SUITABLE FOR USE AND PROVIDE REASONABLE SERVICE LIFE DURING ALL NORMAL OPERATIONS AT TAKE-OFF SPEEDS OF NOT MORE THAN 200 MPH AND LANDING SPEEDS OF NOT MORE THAN 160 MPH.

THE TIRE SHALL WITHSTAND WITHOUT FAILURE THE FOLLOWING TESTS:

TIRE 1:

DYNAMIC TEST - THE TIRE SHALL SATISFACTORILY WITHSTAND 125 CYCLES OF TEST A, 125 CYCLES OF TEST B, AND 60 CYCLES OF TEST C WITHOUT EVIDENCE OF FAILURE.

TEST A - TAKE-OFF. THE TIRE SHALL BE TAXIED ON THE FLYWHEEL FOR 13,500 FEET AT 30 MPH WITH A 38,500 POUND LOAD. THE FLYWHEEL SHALL BE STOPPED, KEEPING THE TIRE FULLY LOADED. THE FLYWHEEL SHALL THEN BE ACCELERATED AT AN AVERAGE RATE OF 4 FT/SEC/SEC TO 200 MPH. THE TIRE SHALL BE UNLANDED AFTER A ROLL DISTANCE OF 10,730 ± 200 FEET HAS BEEN COVERED IN APPROXIMATELY 73 SECONDS. THE LOAD SHALL BE MAINTAINED AT 38,500 POUNDS FOR THE FIRST 15 SECONDS DURING THE TAKE-OFF ROLL. THE LOAD SHALL THEN BE DECREASED LINEARLY WITH TIME DURING THE NEXT 45 SECONDS TO 15,000 POUNDS AND THEN REDUCED TO "0" LOAD AT THE END OF THE TAKE-OFF ROLL.

TEST B - LANDING. THE TIRE SHALL BE LANDED AGAINST THE FLYWHEEL ROTATING AT A SPEED OF 160 MPH WITH AN AVERAGE DECELERATION RATE OF 4.5 FT/SEC/SEC FROM 160 MPH TO 30 MPH. THE LOAD SHALL BE INCREASED LINEARLY DURING THE INITIAL 4 TO 5 SECONDS AFTER LANDING TO 8,000 POUNDS AND THEN INCREASED LINEARLY TO 27,000 LBS 30 SECONDS AFTER LANDING AND MAINTAINED AT THAT VALUE UNTIL THE TIRE HAS DECELERATED TO 30 MPH. THE TIRE SHALL CONTINUE TO ROLL UNTIL A LANDING ROLL DISTANCE OF 5,920 ± 200 FEET HAS BEEN COVERED IN APPROXIMATELY 42.4 SECONDS. THE TIRE SHAL THEN BE TAXIED AT 30 MPH FOR 13,500 FT AT A LOAD OF 27,000 POUNDS.

TEST C - TAXI AND CAMBER. THE FLYWHEEL SHALL BE ACCELERATED TO 30 MPH AND MAINTAINED AT THIS SPEED THROUGHOUT THE TEST. THE TIRE SHALL BE LANDED AGAINST THE FLYWHEEL IN A POSITION OF 10 DEGREE CAMBER. THE LOAD SHALL BE INCREASED LINEARLY IN THE FIRST 1-1/2 SECONDS TO THE FOLLOWING VALUES: 47,000 POUNDS APPLIED LOAD, 46,800 POUNDS VERTICAL LOAD, AND 8,300 POUNDS HORIZONTAL LOAD. IT SHALL REMAIN AT THIS LOAD UNTIL A LANDING ROLL DISTANCE OF 2,300 FEET HAS BEEN COVERED. THERE SHALL BE 30 OF THE ABOVE CYCLES COMPLETED AT 10 DEGREE CAMBER OUTBOARD AND 30 CYCLES OF 10 DEGREE CAMBER INBOARD. SWING CARRIAGE SHALL BE USED. IF THIS CAPABILITY DOES NOT EXIST, AN ACCEPTABLE ALTERNATE IS THE FIXED LOAD CARRIAGE "BENT MANDREL" PROCEDURE. IF THE BENT MANDREL IS USED THE CAMBER TEST SHALL BE CONDUCTED AS ABOVE WITH THE EXCEPTION THAT THE APPLIED LOAD SHALL BE 48,000 LBS.

TIRE 2:

BURST TEST - THIS TIRE SHALL BE SUBJECTED TO A HYDROSTATIC BURST TEST AND THE FAILING PRESSURE, DESCRIPTION OF FAILURE AND LOCATION SHALL BE REPORTED IN THE QUALIFICATION TEST REPORT.

TIRE 3:

GOVT LAB TEST - THIS TIRE SHALL BE RETREADED FROM THE SAME BATCH AS TEST TIRE 1 AND 2 AND SHALL BE IDENTICALLY PROCESSED. THIS TIRE SHALL NOT BE CONTRACTOR TESTED AND SHALL BE PROVIDED FOR GOVERNMENT LABORATORY TESTING.

SECTION - A REPRESENTATIVE SECTION OF TIRE 1 AND TIRE 3 SHALL BE SUBMITTED TO A GOVERNMENT LABORATORY AS SPECIFIED IN MS 3377(AS).

AIR RETENTION - THE TUBELESS TIRE SHALL BE INFLATED TO A PRESSURE OF 150 PSI AND ALLOWED TO STAND FOR A PERIOD OF 4 HOURS AT WHICH TIME THE PRESSURE DROP DUE TO GROWTH SHALL BE REPLACED. THE TIRE SHALL THEN STAND FOR AN ADDITIONAL 12 HOURS AT WHICH TIME THE PRESSURE SHALL BE MEASURED AND THE TIRE INSPECTED. THE AIR PRESSURE LOSS SHALL NOT EXCEED 5 PERCENT AND THE TIRE SHALL NOT REVEAL ANY APPEARANCE OR PERFORMANCE DEFECTS, SUCH AS SIDEWALL BLISTERS, TREAD SEPARATION, ETC.

NOTES - REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

(A) CANCELLED AFTER 11 JUNE 1981 NO SUPERSEDING STANDARD.

APPROVED 18 FEB 1969 REVISED (A) 11 JUNE 1981

This standard has been approved by the Naval Air Systems Command Department of the Navy and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. AS Other Cust	TITLE RETREAD TIRE - AIRCRAFT 20.00-20 (200 MPH) TYPE II (NAVY)	MILITARY STANDARD MS 3389(AS)
PROCUREMENT SPECIFICATION MIL-R-7726	SUPERSEDES:	SHEET 1 OF 1

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